1. Project Objective

Build a layered Data Warehouse solution that integrates CRM and ERP data for reporting and analytics using bronze, silver, and gold layers.

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2. Scope

The project will include:

- Data ingestion from source systems
- Data cleansing and transformation
- Star schema data modelling
- BI-ready datasets
- Reporting integration

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3. Architecture Overview

Sources → Bronze Layer → Silver Layer → Gold Layer → Power BI

Layers

Layer	Purpose
Bronze	Raw source data
Silver	Cleaned and standardized data
Gold	Business-ready analytical data

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4. Project Phases

Phase	Activities
Planning	Analysed requirements and define scope
Architecture Design	Design data flow and warehouse layers
Initialization	Setup database, schemas, and Git repository
Bronze Layer	Load raw CRM and ERP data

Phase	Activities
Silver Layer	Clean and transform data
Gold Layer	Build star schema and fact/dimension tables
Testing	Validate data and ETL processes
Reporting	Connect Power BI dashboards
Deployment	Deploy and schedule ETL jobs

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5. Main Tables

Bronze & Silver Tables

- crm_sales_details
- crm_cust_info
- crm_prd_info
- erp_cust_az12
- erp_loc_a101
- erp_px_cat_g1v2

Gold Tables

- fact_sales
- dim_customers
- dim_product

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6. Technology Stack

Component	Tool
Database	SQL Server
ETL	SQL Stored Procedures
Version Control	Git/GitHub
Visualization	Power BI

Component	Tool
Documentation	Notion
Diagramming	Draw.io

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7. Deliverables

- Data warehouse architecture
- SQL scripts
- ETL stored procedures
- Star schema model
- Power BI dashboards
- Technical documentation

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8. Expected Outcome

A fully functional Data Warehouse solution that supports business reporting, analytics, and decision-making.